



A Collaborative Framework for Evaluation of Run-Time Changes in Enterprise Web Services

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ABSTRACT

Increasing market demand and technology in today's world has made change management an inevitable one. To remain competitive, organizations must be able to respond to change in a timely and cost effective manner. Although there are many change management issues and solutions, there is no proper support for efficient change evaluation and monitoring. Present change management works are done only in the web service interface such as WSDL Web Service Definition Language and there is no proper support for dynamic nature in them. In this paper, we propose a novel dependency analysis approach for evaluating the changes specified by the analyst in an effective manner and a FSM based model is utilized for exhibiting the dynamic nature. This measure of change evaluation ensures that the business analyst has a direct control over the changes he makes and provides an environment that notifies the analyst about the outcome in a meaningful way. We propose a set of five change factors based on which the dependency existing between the business rules, functions and parameters is analyzed. These change factors are also involved in the change evaluation that makes the changes to be specified in a precise and formal manner. We develop a prototype system of the proposed model to demonstrate its effectiveness. We also conduct an experimental study to assess the performance of the change management along with the change measure as an output.

Categories and Subject Descriptors

H.4 [Information Systems Applications]: Office Automation – Workflow management

General Terms

Management, Measurement, Documentation, Performance, Design, Economics, Reliability, Experimentation, Human Factors, Standardization, Languages, Theory, Legal Aspects, Verification.

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Keywords

Web Services Change Management; Enterprise Business Scenario; Dependency Analysis; Similarity Measure

1. INTRODUCTION

A web service is a piece of business logic that enables a paradigm shift in business structures allowing them to outsource required functionality from third party Web-based providers. A consumer of a web service is not tied to the service directly, as the service interface can change over time without compromising the client's ability to interact with it. As services are not static, they undergo many changes during their life cycle. Hence there arises the need for an effective framework managing those changes without affecting the business functionality. Change management is the process responsible for controlling and managing the lifecycle of all changes in an IT environment. Change management is critical and challenging in the development and maintenance of service-based applications and information systems due to the distributed and dynamic nature of services. Change management is defined as the process of controlling changes to the infrastructure or any aspect of services, in a controlled manner, enabling approved changes with minimal disruption. Most companies relying on an Information Technology (IT) system for their daily operations heavily invest in its maintenance. Tools that monitor network traffic, record anomalies and keep track of the changes that occur in the system are usually used. Every business organization is under intense pressure to deal with accelerated business cycles. An organization's success depends upon its agility for responding quickly to today's complex, ever-changing markets and regulatory climate. For example, a company's ability to implement new pricing models or change product or service features to meet or beat the competition is a key differentiator in ensuring business success. But many organizations cannot react quickly to business changes. Their key business policies and regulatory compliance information is locked inside multiple software systems, expressed in highly technical languages and generally inaccessible to the business managers responsible for implementing business policies and rules. Businesses today need a solution that supports business changes and cycles far more effectively than traditional methods, and takes direct advantage of business expertise.

Change impact analysis and change evaluation are two main activities involved in the management of IT systems. Currently, there exists no universal model to guide analysts while performing these activities. Although the Information Technology

Infrastructure Library (ITIL) provides a guide to the organization and structure of the tools and processes used to manage IT systems, it does not provide any models that can be used to implement the required features. ITIL emphasizes the need to identify other CIs that will be impacted whenever a change to a specific CI is proposed. Change management is responsible for managing changes to both hardware and software components. These changes may be a solution to a previous incident or problem, or may be a proactive measure to reduce cost or improve performance. Change management is an IT service management discipline. The objective of change management in this context is to ensure that standardized methods and procedures are used for efficient and prompt handling of all changes to control IT infrastructure, in order to minimize the number and impact of any related incidents upon service. Changes in the IT infrastructure may arise reactively in response to problems or externally imposed requirements, e.g. legislative changes, or proactively from seeking improved efficiency and effectiveness or to enable or reflect business initiatives, or from programs, projects or service improvement initiatives. Change Management can ensure standardized methods, processes and procedures which are used for all changes, facilitate efficient and prompt handling of all changes, and maintain the proper balance between the need for change and the potential detrimental impact of changes. Hence there arises the need for a universal model that can assist the analyst in analyzing the impacts made by the changes specified by him.

In our work, we propose a dependency analysis model for analyzing and evaluating the impact made by the changes in a service. This measure of change evaluation ensures that the business analyst has a direct control over the changes he makes and provides an environment that notifies the analyst about the outcome in a meaningful way. We propose a set of five change factors based on which the dependency existing between the business rules, functions and parameters is analyzed. These change factors are also involved in the change evaluation that makes the changes to be specified in a precise and formal manner. We develop a prototype system of the proposed model to demonstrate its effectiveness. We also conduct an experimental study to assess the performance of the change management along with the change measure as an output.

2. RELATED WORKS

In this section, we discuss the research and efforts that are closely related to our work. Specifically, this section deals with work related to Web services, workflow change management and Petri-nets.

Yang et al presented a method for detecting failures in implementation of services that are running currently in concurrent order and also from rules in Service Oriented architecture based atmospheres. It also finds the dependencies that co exist among the services and are extracted from log files when any change occurs [1]. Business dealings are obligatory to accomplish in the company of changes within legislation and strategy. However the successful executions of innovative legislation are a lot expensive, might include long escort period which cause huge loss to industry trade. Yiwei Gong and others proposed a novel technique for business processes in favor of building liveness and agility while applying changes through innovative or improvised policies. It also elaborated a new research method for software reuse by joint policy execution [2].

Dimitris Apostolou et al presented an ontology driven method for managing changes in e-Government services based on SOA. It enables methodical reaction of these systems to changes by applying proper techniques intended for accomplishing stability whenever a change is exposed [3]. In [4], the authors presented a novel method to represent changes in ontologies which have been organized in layers and provided data independence from versions, etc. Also deposits of change circulation stratagems are allowed to keep distributed copy of the same ontology synchronized. A methodology for handling changes in Long-term Composed Services (LCS) was proposed by Xumin Liu, Athman Bouguettaya and others. In this technique the authors have categorized changes in two ways:- Top-down and Bottom-Up scaffolds. Also the functional and non-functional properties of the composed services have been identified [5], [6], [14], [16], [20], [21], and [22]. As business policies change continuously, there is a need for service-oriented systems to be submissive and adaptive. Hans Weigand et al presented an elucidation to policy changes in business atmospheres by dividing the SOA based architectures on policy enforcements and their specifications [7]. Their method has been highly successful to Adaptive Service Oriented Architectures (ASOA). H. Yahyaoui and others formulated a method to manage and handle the changes propagated at run time through Web Service Policy Language (WSPL) which permits perspective pattern by mutually policy and rule stages [8]. Pedro Antunes and Hernâni Mourão throw in Business Process Management systems (BPM) through resilience sustain. The system introduced a deposit of services integrating resilience support in BPM. Harry Jiannan Wang and J. Leon Zhao modeled a new approach for changes in work flow management systems called Constraint-centric Workflow Change Analytics (CWCA). The core functionality of this system is it analyzes change anomalies in workflow based architectures. An algorithm (procedure) is used to discover changes in SOA context and open-source rule engine is exposed to afford a verification of notion execution of CWCA [10]. Shari S.C. Shang, Eldon Y. Li, Ya-Ling Wu, Oliver C.L. Hou have done a survey on existing life cycle models in context of web services. These representations illustrate the variety of open Web 2.0 relevance and presented a scaffold in support of a superior kind of working patterns and significance scheme in the Web 2.0 model [11]. Minhong Wang and Huaqing Wang presented a model for managing business logic of the services within the business processes. Since many of the existing models cope up with changes at process level, this paper has identified the changes propagated at business layer [12]. Daniel Żmuda, Marek Psiuk and Krzysztof Zieliński presented some innovative challenges in SOA based environments to observe and watch the implementation strategies occurring dynamically. Dynamic run-time reconfiguration is the prime concept of that approach [13].

2.1 Limitations in the Related Works Section

Though the existing system satisfies the change management issues to a certain extent, there are certain limitations and issues that the existing system should satisfy. They are as follows:

- The existing system does not provide complete visualization of the logic to the analyst.
- Only the structural and semantic verification of the change is analyzed.
- Supports changes only in the service definitions such as WSDL rather than schema.

- Lack of dynamism and runtime management of changes.
- No proper support for the change specification and change evaluation.

3. PROTOTYPE OF THE SYSTEM

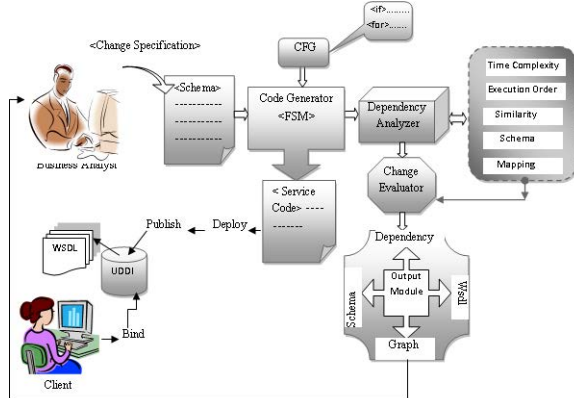


Figure 1. Overall architecture of the system.

The architecture of the proposed system is shown in Figure 1. It consists of the following components such as service code generator, dependency analyzer, change evaluator and output module. The Analyst first specifies the changes in a formal way through the schema. This schema is based on the context of the business logic where the tags are in the form of rules, functions and parameters. This schema actually depicts the service code in a more formal way with the tags and attributes. After the specification and modification in the schema, the service code gets modified and gets build by the code generator.

The code generator is one of the major components, which acts as an intermediate between the schema and the code. The code generator actually parses the schema and converts into BL tags containing the rules, functions and parameters. The grammar for the code logic such as the control and loop statements are provided by the CFG Context free Grammar. Based on the CFG and the logic the service code gets generated. This code generation is performed within a considerable amount of time. Apart from these, the code generator is concerned with the verification of the correctness with the generated code. After the service code generation, the service code is deployed automatically and registered in the UDDI for further usage by the client.

Dependency Analyzer discovers dependencies with the ascertained rules and functions in the business logic through the identified change criteria. The change factors are updated frequently with respect to the changes performed by the analyst each time and they ascertain whether the changes performed are finite and valid. Based on the dependency analysis, the analyst is capable to ascertain the changes performed by him.

Change Evaluator evaluates and measures the changes based on the changes updated in the dependency analysis. This measure is to enable the analyst to understand the logic in a more comfortable way. The range to which the rules and functions varies with respect to the changes is measured and reported by this component.

The output module is concerned with submitting the reports to the analyst for each and every change performed. These reports are in the form of graph, schema, and WSDL and dependency measure. The graph provides the growth rate analysis of the logic for every change. The schema is concerned with providing an overview of the changes in schematic format. The WSDL file is the file generated dynamically for each change in the service code. The dependency also forms an output measure for the analyst.

4. IMPLEMENTATION

4.1 Overview

We implemented the system to provide a practical study of the proposed method. We use the airline reservation system and credit card service in our running example as the scenario. The purpose of the implementation is twofold. First, we implemented the proposed change management algorithms to demonstrate their practicality. The system provides a graphic user interface for an analyst to submit a change specification in the form of schema as in figure 2 and figure 3. After specification, it automatically generates the process of change analysis. The corresponding code for the schema is generated automatically for every changes specified by the analyst in the schema as shown in figure 4. Second, we conducted a set of experiments to assess the performance of the service for every change.

Once an analyst submits a change specification, the system generates the new logic code within a reasonable time range. We identified a set of parameters that affect the performance of the service such as cost of a ticket and availability of tickets in an airline reservation system. Apart from this analysis, the dependency between the services is shown in terms of similarity, execution order, mapping function and schema. Finally the WSDL file with respect to the changes is generated.

The system provides a convenient interface that enables an analyst to specify a change in a declarative way through schema. After this step, the proposed system automatically implements the change and generates the new logic code that satisfies the request and change requirement. The system provides a visualization component to see the intermediate change steps. This enables the analyst to perform change analysis during the change management process.

4.2 Snapshots

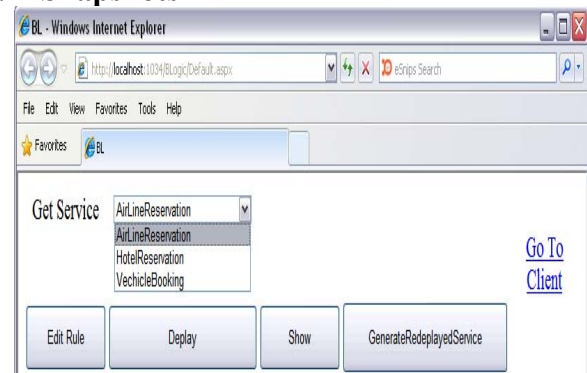


Figure 2. Interface for Business Analyst.

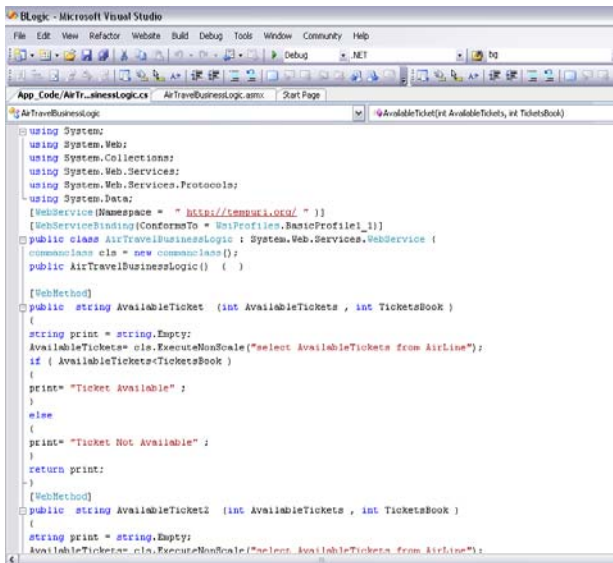


Figure 3. Source Logic.

```

<?xml version="1.0" encoding="utf-8" ?>
<AmountValidation>
  <Route function="AvailableTicket" >
    <para type="int" name="AvailableTickets" , "></para>
    <para type="int" name="TicketsBook" ></para>
    <if condition="AvailableTickets<TicketsBook">
      <process>
        <output_statement>
          <print>"Ticket Available"</print>
        </output_statement>
      </process>
    </if>
    <else>
      <process>
        <output_statement>
          <print>"Ticket Not Available"</print>
        </output_statement>
      </process>
      <return>Yes</return>
    </else>
  </Route>
  <Route function="AvailableTicket2" >...
</AmountValidation>

```

Figure 4. Business Logic Schema.

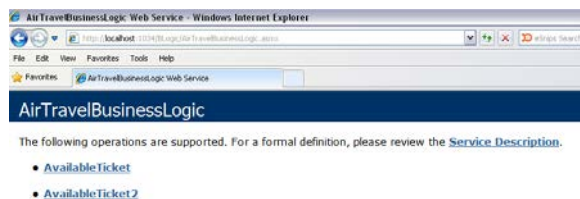


Figure 5. Redeployed Web Service after incorporating changes.

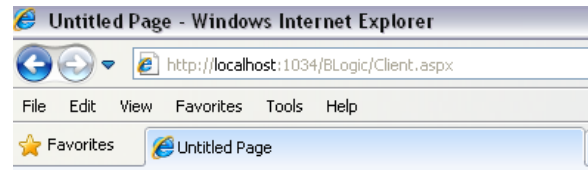


Figure 6. Client consuming the Web Service.

4.3 Performance Study

We conducted a set of experiments to assess the performance of the change management algorithm. We use the airline service example as our testing environment to setup the experiment parameters. We considered the two change parameters cost, availability of tickets and analyzed their effect on the service performance. Apart from this, we analyze the five change factors such as execution order, time complexity, mapping function, similarity and schema. We run our experiments on a Windows 7 Operating System with an Intel Core i5 processor, 4-GB of RAM.

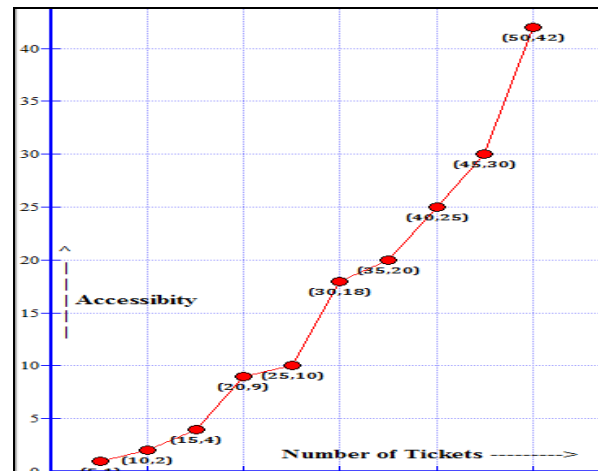


Figure 7. Tickets v/s. Accessibility.

The figure 7 describes the change curve corresponding to the changes in the number of tickets. From the graph, it is clear that the accessibility of the service is increased with increase in the number of tickets offered by the service providers. Thus for each change made, the analyst can view the change curve and assures the effect of changes.

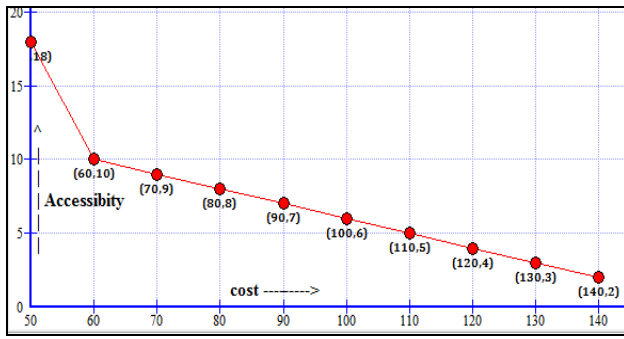


Figure 8. Cost v/s. Accessibility.

The figure 8 is similar to the figure 7 except that the parameter 'cost' is used for verifying the accessibility of the service. From the graph, it is clear that the accessibility is decreased with increase in the cost of tickets.

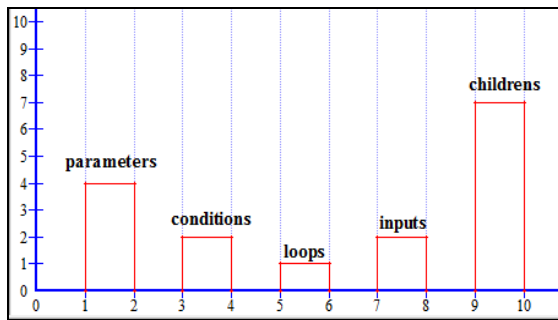


Figure 9. Schema Dependency.

Apart from the performance measurement, the effect of changes in the change factor is also analyzed. Figure 9 shows the variation of the schema and the constructs for every change in the logic.

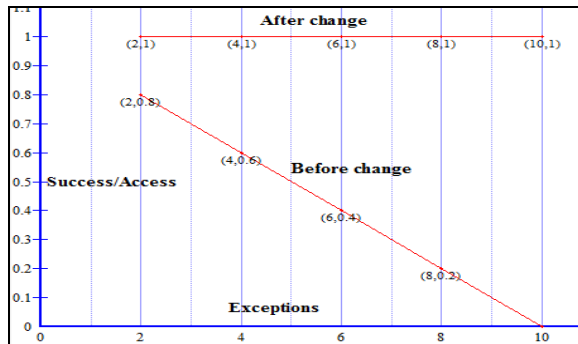


Figure 10. Change Measure in Execution Order.

The Figure 10 shows a sample graph with respect to change in the execution order of the logic. We took car rental service composed with airline reservation system for our example. The sequence of execution of service before change and after change is as follows:

Normal:

Login -Book_ticket – car rental – credit_card

After change:

Login - Book_ticket – credit_card - car rental

The success/access ratio for car rental service is calculated for various exception rates such as invalid credit card. From the Figure 10, it is clear that the success/access ratio remains constants after change for all invalid credit card exceptions since the car rental service is not further accessed, if the credit card validation service fails there by reducing the overhead in the car rental service.

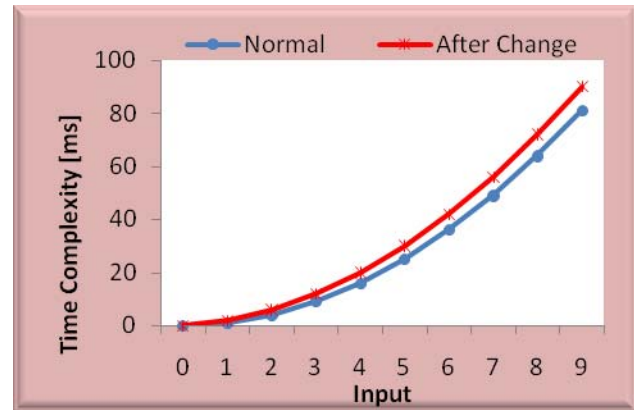


Figure 11. Time Complexity.

The figure 11 shows the effect of changes on the time complexity of the logic. Corresponding to the change [addition or deletion] of the logic, the time complexity increases or decreases from the normal one for various inputs. This change is measured and plotted in the graph for assisting the analyst in performing future changes.

The mapping function comes under the category of reducibility of the service where one service is reduced into another without any change in the input it receives and output it produces. In case of mapping, only the processing logic is changed without modifying the service functionality. In our example, we first check and validate the username and password length. Then, we restrict ourselves to the password length validation alone. This change does produce the same output as before for certain inputs to the login.

5. CONCLUSION

The proposed model provides a powerful platform to enable changes in the service business logic more sophisticatedly in such a way, that the interoperability between the services is managed with flexibility and ease. This paper identifies the various dependency properties needed for managing changes in the business logic. The model is capable of specifying, react to and evaluate changes initiated by the analyst. We then proposed a set of change factors for managing changes and their evaluation. We first defined the various change factors and derived a mathematical solution for each one. The derived solution is considered as the change measure for individual change factor. We then implemented our model using the Airline Reservation System as a practical example and evaluated the various change factors involved in it. Finally, the performance of the system is

analyzed, in which, the changes are measured and reported in the form of a graph. Change management in service oriented enterprises presents vast opportunities for research. This work attempts to initiate work in this field. Future work may include features such as growth rate analysis of changes where this methodology can be migrated to cellular automata, where each cell has certain business rules pertaining to that cell and the neighboring cells can be mapped using this dependency measure.

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